STAT 5200, Notes 10[1]

Minchul Shin[2]
Fall 2024

Readings. Finish reading Chapter 5 of W, and read Chapter 6. These notes cover Sections 5.2.6 and 5.3, and Section 6.1.

Difficulties posed by 2SLS estimation. There are two topics to discuss, the possible bias of the 2SLS estimator, and the magnitudes of the standard errors of the estimates.

Bias. Substantial bias can occur in 2SLS estimation if the instruments are weak. Let's consider the simple model

$$y = \beta_0 + \beta_1 x_1 + u, \tag{1}$$

where the disturbance term u has mean zero. Assume x_1 is endogenous and let z_1 be the instrument for x_1. Then the estimator of $\beta = (\beta_0, \beta_1)'$ is given by (23) of Notes 8 [use only the first line of (23)],

$$\widehat{\beta}_{IV} = \beta + \left[\left(\sum_{i=1}^{N} \mathbf{x}_i'\mathbf{z}_i\right)\left(\sum_{i=1}^{N} \mathbf{z}_i'\mathbf{z}_i\right)^{-1}\left(\sum_{i=1}^{N} \mathbf{z}_i'\mathbf{x}_i\right)\right]^{-1}\left(\sum_{i=1}^{N} \mathbf{x}_i'\mathbf{z}_i\right)\left(\sum_{i=1}^{N} \mathbf{z}_i'\mathbf{z}_i\right)^{-1}\left(\sum_{i=1}^{N} \mathbf{z}_i' u_i\right) \tag{2}$$

where $\mathbf{x}_i = (1, x_{i1})'$ and $\mathbf{z}_i = (1, z_{i1})'$. Asymptotically (2) is

$$\text{plim}\,\widehat{\beta}_{IV} = \beta + \left[E[\mathbf{x}'\mathbf{z}]\left(E[\mathbf{z}'\mathbf{z}]\right)^{-1}E[\mathbf{z}'\mathbf{x}]\right]^{-1}E[\mathbf{x}'\mathbf{z}]\left(E[\mathbf{z}'\mathbf{z}]\right)^{-1}E[\mathbf{z}'u]. \tag{3}$$

All the 2×2 matrices in (3) are invertible, and thus

$$\begin{aligned}
\text{plim}\,\widehat{\beta}_{IV} &= \beta + \left(E[\mathbf{z}'\mathbf{x}]\right)^{-1}E[\mathbf{z}'u] \\
&= \beta + \begin{pmatrix} 1 & E[x_1] \\ E[z_1] & E[z_1 x_1] \end{pmatrix}^{-1}\begin{pmatrix} E[u] \\ E[z_1 u] \end{pmatrix} \\
&= \beta + \frac{1}{Cov(z_1, x_1)}\begin{pmatrix} E[z_1 x_1] & -E[x_1] \\ -E[z_1] & 1 \end{pmatrix}\begin{pmatrix} 0 \\ Cov(z_1, u) \end{pmatrix}.
\end{aligned} \tag{4}$$

Then

$$\text{plim}\,\widehat{\beta}_{IV,1} = \beta_1 + \frac{Cov(z_1, u)}{Cov(z_1, x_1)} \tag{5}$$

[1] This is based on Paul Shaman's lecture notes (Notes 10 from STAT 5200) at the University of Pennsylvania. Last revised: Summer, 2024. Last compiled: August 26, 2024.

[2] Email: mincshin@wharton.upenn.edu

and

$$\text{plim}\,\widehat{\beta}_{IV,0} = \beta_0 - E[x_1]\frac{Cov(z_1,u)}{Cov(z_1,x_1)}. \tag{6}$$

Thus, if the instrument z_1 has some correlation with u, the 2SLS estimator of β is asymptotically biased. Moreover, if the instrument is weakly correlated with the endogenous explanatory variable, the bias can be substantial. Thus, even if the correlation between z_1 and u is modest, a weak instrument can cause large bias.

Analogous to (5), for the OLS estimator of the slope, we have

$$\text{plim}\,\widehat{\beta}_{OLS,1} = \beta_1 + \frac{Cov(x_1,u)}{Var(x_1)}. \tag{7}$$

From (5) and (7) we have

$$\frac{\text{plim}\,\widehat{\beta}_{IV,1} - \beta_1}{\text{plim}\,\widehat{\beta}_{OLS,1} - \beta_1} = \frac{Cov(z_1,u)/Cov(x_1,u)}{Cov(z_1,x_1)/Var(x_1)}. \tag{8}$$

And, in fact, the same expression holds for the ratio of the two asymptotic bias expressions for estimation of the intercept. If the instrument has some correlation with u, it is clear from this expression that the 2SLS estimator can sometimes have less bias, and sometimes more bias, than the OLS estimator in large samples.

Finite-sample bias. Recall that 2SLS estimator is an IV estimator with a particular choice of the instrument, $\widehat{x}_1$, which is based on the first stage. In the current simple example without other covariates, the first stage can be written as

$$x_1 = \theta_0 + \theta_1 z_1 + ... + \theta_M z_M + r_1.$$

In a matrix form, we can write it as

$$X_1 = Z\theta + R_1$$

where $\theta = [\theta_0, \theta_1, ..., \theta_M]'$. Then,

$$\widehat{X}_1 = Z\widehat{\theta} \quad \text{where} \quad \widehat{\theta} = (Z'Z)^{-1}(Z'X_1)$$

In finite sample, $\widehat{X}_1$ is a somehow related with X_1, which in turn creates an unwanted correlation between $\widehat{x}_1$ and u (because x_1 and u are correlated).

Now, let us examine 2SLS estimator

$$\widehat{\beta}_{2SLS,1} = \beta_1 + \frac{\widehat{cov}(\widehat{x}_1,u)}{\widehat{cov}(\widehat{x}_1,x)}$$

Note that bias can arise even in finite sample with 2SLS estimation,

$$bias = E\left[\widehat{\beta}_{2SLS,1} - \beta_1\right] = E\left[\frac{\widehat{cov}(\widehat{x}_1, u)}{\widehat{cov}(\widehat{x}_1, x)}\right]$$

where the numerator may not be zero when $M > 1$ for the reason explained above. The denominator measures the strength of instruments. When instruments are weak, the bias of the 2SLS estimator gets larger.

Standard error and variance inflation factor (VIF). The variance inflation factor is discussed in the Appendix to Notes 4. We revisit it here.

Consider the univariate regression setting (1), and assume x_1 is exogenous, and, as above, let $\mathbf{x} = (1, x_1)'$. The standard error of the OLS estimate of β_1 is given by the square root of the second diagonal entry of the matrix

$$\widehat{\sigma}^2 \begin{pmatrix} N & \sum x_{i1} \\ \sum x_{i1} & \sum x_{i1}^2 \end{pmatrix},$$

or the square root of

$$\frac{\widehat{\sigma}^2}{\sum_{i=1}^{N}(x_{i1} - \overline{x}_1)^2}, \tag{9}$$

where $\widehat{\sigma}^2$ is the sum of squared residuals from the OLS fit, divided by $N - 2$. The denominator can be interpreted as a sum of the squared residual that is from the regression of x_{i1} on all other explanatory variables, which is an intercept in this simple case. Note that $\overline{x}_1$ can be interpreted as the prediction based on the projection of x_1 on 1.

Now we extend above based on the multiple regression

$$y = \gamma_0 + \gamma_1 x_1 + \gamma_2 x_2 + \cdots + \gamma_K x_K + v. \tag{10}$$

Consider the multiple regression setting (10), and focus on estimation of γ_K. Assume all the explanatory variables are exogenous. Apply (2.56) of Property LP.7 in Wooldridge: the OLS estimator of γ_K in (10) is obtained by performing a simple regression, projecting y onto 1, $\widetilde{x}_K$, where $\widetilde{x}_K = x_K - L(x_K|1, x_1, ..., x_{K-1})$ is the residual from the multiple regression of x_K onto $1, x_1, ..., x_{K-1}$. Denote the R square from this multiple regression by $\widehat{R}^2_{K\cdot 2,...,K-1}$. It is given by

$$\widehat{R}^2_{K\cdot 1,...,K-1} = 1 - \frac{\sum_{i=1}^{N} \widetilde{x}_{iK}^2}{\sum_{i=1}^{N}(x_{iK} - \overline{x}_K)^2} = 1 - \frac{\widetilde{SSR}_K}{\widetilde{SST}_K}. \tag{11}$$

That is, $\widetilde{SSR}_K$ denotes the sum of squared residuals from the multiple regression of x_K onto $1, x_1, ..., x_{K-1}$, and $\widetilde{SST}_K$ denotes the total sum of squares of x_K. It follows from (11) that we can write

$$\widetilde{SSR}_K = (1 - \widehat{R}^2_{K\cdot 1,...,K-1})\widetilde{SST}_K. \tag{12}$$

By (9) it follows that the standard error of the OLS estimate of γ_K, the partial slope attached to x_K in (10), is the square root of

$$\frac{\widetilde{\sigma}^2}{\widetilde{SSR}_K} \tag{13}$$

where $\widetilde{\sigma}^2$ is the sum of squared residuals from the multiple regression fit, divided by $N-K-1$:

$$\widetilde{\sigma}^2 = \frac{1}{N-K-1}\sum_{i=1}^{N}(y_i - \widehat{\gamma}_0 - \widehat{\gamma}_1 x_1 - \widehat{\gamma}_2 x_2 - \cdots - \widehat{\gamma}_K x_K)^2,$$

and the denominator is the sum of the squared residual that is from the regression of x_{iK} on all other explanatory variables, $1, x_{1i}, ..., x_{K-1i}$ in this case.

Using (12), we can write the squared standard error for $\widehat{\gamma}_K$ as

$$\frac{\widetilde{\sigma}^2}{\widetilde{SSR}_K} = \frac{\widetilde{\sigma}^2}{(1-\widehat{R}^2_{K\cdot 1,...,K-1})\widetilde{SST}_K} = \frac{\widetilde{\sigma}^2}{\widetilde{SST}_K} VIF_K. \tag{14}$$

Here

$$VIF_K = \frac{1}{1-\widehat{R}^2_{K\cdot 1,...,K-1}} \tag{15}$$

is the *variance inflation factor* (VIF) for OLS estimation of γ_K attributable to the correlation of x_K with the other explanatory variables collectively. All else equal, a higher correlation leads to a larger standard error.

There is a variance inflation factor for each partial slope in the model (10). A variance inflation factor is greater than or equal to 1 and is equal to 1 only if the explanatory variable is uncorrelated (in the sample) with all of the other explanatory variables. Thus, collinearity inflates standard errors and tends to deflate t and F statistics. The stronger the collinearity involving an explanatory variable, the greater is the VIF for that variable.

The extension of (14) to 2SLS is straightforward. Suppose we are doing 2SLS estimation for the model (10). Recall (26) of Notes 8, giving the estimator of the asymptotic covariance matrix of the 2SLS estimator of the parameter vector,

$$\widehat{\sigma}^2\left(\sum_{i=1}^{N}\widehat{\mathbf{x}}_i\widehat{\mathbf{x}}_i\right)^{-1} = \widehat{\sigma}^2(\widehat{\mathbf{X}}'\widehat{\mathbf{X}})^{-1}.$$

Recall also that $\widehat{\mathbf{x}} = (1, \widehat{x}_1, \widehat{x}_2, ..., \widehat{x}_K)'$ is the input to the second-stage regression. In this vector $\widehat{\mathbf{x}}$, the element $\widehat{x}_i = x_i$ if x_i is exogenous, and $\widehat{x}_i$ is the prediction of x_i from the regression of x_i on $\mathbf{z}$ if x_i is endogenous. By analogy with (14) (and with a slight change of notation, to conform to Wooldridge), the standard error of the 2SLS estimator of γ_K is the square root of

$$\frac{\widehat{\sigma}^2}{\widehat{SSR}_K} = \frac{\widehat{\sigma}^2}{(1-\widehat{R}^2_{K\cdot 1,2,...,K-1})\widehat{SST}_K}, \tag{16}$$

where $\widehat{\sigma}^2$ is the 2SLS estimate of the variance of the disturbance term v in (10), $\widehat{SSR}_K$ is the sum of squared residuals from the regression of $\widehat{x}_K$ on $1, \widehat{x}_1, ..., \widehat{x}_{K-1}$, $\widehat{SST}_K$ is the total variation of $\widehat{x}_K$ in the sample, and $\widehat{R}^2_{K\cdot 1,...,K-1}$ is the R square from the regression of $\widehat{x}_K$ on $1, \widehat{x}_1, ..., \widehat{x}_{K-1}$. $\widehat{SST}_K$ is calculated as

$$\widehat{SST}_K = \sum_{i=1}^{N} \left(\widehat{x}_{iK} - \overline{\widehat{x}}_K \right)^2. \tag{17}$$

From (16) we can conclude that collinearity can inflate standard errors in 2SLS regression. Another way of stating this: If the instruments are weak, $\widehat{R}^2_{K\cdot 1,...,K-1}$ can be close to 1, leading to considerable inflation. See Wooldridge's discussion below (5.44).

The standard error can also be made large if the total variation of $\widehat{x}_K$, $\widehat{SST}_K$, is small in the sample. More precisely, what is $\widehat{SST}_K$? If x_K is exogenous, it is just the total variation of x_K in the sample. Thus, in this case, the quality of the instruments has no impact upon the size of $\widehat{SST}_K$. If x_K is endogenous, however, $\widehat{SST}_K$, the total variation of $\widehat{x}_K$ in the sample, can be small if x_K is weakly related to the instruments, leading to a large standard error in (16). That is, if x_K is weakly related to the instruments, $\widehat{SST}_K$ can be close to zero. It is common in 2SLS estimation that coefficients attached to exogenous explanatory variables are estimated more precisely than coefficients attached to endogenous explanatory variables.

Wooldridge notes that weak instruments can lead to very large VIF values, greatly inflating standard errors. See his discussion in Section 5.2.6.

For this whole discussion we also need to consider sample size. Standard errors become smaller as sample size increases. As long as the instruments have some correlation with the endogenous variables, the standard errors will decrease as the sample size increases. How large the sample size needs to be to get "suitable" standard errors clearly depends on context. Research has shown, though, that 2SLS with weak instruments can produce substantial bias, even for large sample sizes. And sometimes a large number of weak instruments can exacerbate the bias problems.

Detecting weak instruments. Conceptually, detecting weak instruments is easy. We can perform a test based on the first-stage equation. Consider a simple situation that x_K is the only endogenous variable. The first-stage equation is then would look like

$$x_K = \delta_0 + \delta_1 x_1 + ... + \delta_{K-1} x_{K-1} + \theta_1 z_1 + ... + \theta_M z_M + r_K.$$

We can test the following hypothesis

$$H_0 : \theta_1 = \theta_2 = ... = \theta_M = 0$$

Note that there are M number of restrictions. We can construct a F-statistic (or, Wald statistic). The idea is to compare the unrestricted model:

$$x_K = \delta_0 + \delta_1 x_1 + ... + \delta_{K-1} x_{K-1} + \theta_1 z_1 + ... + \theta_M z_M + r_K,$$

to the restricted model:

$$x_K = \delta_0 + \delta_1 x_1 + ... + \delta_{K-1} x_{K-1} + r_K^*.$$

If we reject the hypothesis (or, the test statistic is larger than the pre-specified critical value; the rule-of-thumb critical value is "10"; see Staiger and Stock (1997)), then we may conclude that the instruments at hand are not jointly weak.

If we cannot reject the hypothesis (or, the test statistic is less than the pre-specified critical value), then we should be concerned with weak instruments.

Summary of the recent literature:

- When there one or multiple endogenous variable(s) and homoscedastic first-stage:
  - A F-testing procedure described above (or, a test based on the similar idea) is valid. See the following papers:

    Staiger, D.O. and Stock, J.H., 1994. Instrumental variables regression with weak instruments.

    Stock, J. and Yogo, M., 2005. Asymptotic distributions of instrumental variables statistics with many instruments. Identification and inference for econometric models: Essays in honor of Thomas Rothenberg, 6, pp.109-120.

- When there is one endogenous variable and non-homoscedastic first-stage:
  - Use the Montiel Olea and Pflueger (2013) effective first-stage F-statistic. See the following paper:

    Olea, J.L.M. and Pflueger, C., 2013. A robust test for weak instruments. Journal of Business & Economic Statistics, 31(3), pp.358-369.

- When there are multiple endogenous variable and non-homoscedastic first-stage:
  - It has been an open question, and it seems that the authors of the following working paper solved it:

    Lewis, D.J. and Mertens, K., 2022. A robust test for weak instruments with multiple endogenous regressors.

What should we do if instruments are weak? Researchers discovered that asymptotic inference based on 2SLS that we developed in this course breaks down when instruments are weak.

It turns out that the valid asymptotic inference is possible even under such situation. Such inferential procedure is in general called "*weak identification robust* procedure." It is beyond the scope of this course. If you are interested in, the following survey paper is a good starting point:

- Andrews, I., Stock, J.H. and Sun, L., 2019. Weak instruments in instrumental variables regression: Theory and practice. Annual Review of Economics, 11, pp.727-753.